IGCSE 0607 Practice Questions — E1.10

Ratio and proportion

Total Questions: 18 — **Total Marks:** 105 Marks

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1 **(a)** Write 45 : 75 : 105 in its simplest form.

..... **[1]**

(b) Alice, Bob and Charlie share \$240 in the ratio of their ages. Alice is 8 years old, Bob is 10 years old and Charlie is 12 years old.
Work out how much Bob receives.

% ..... **[2]**

2 **(a)** Write the ratio 1.5 hours to 45 minutes in its simplest form.

..... **[1]**

(b) Alex, Ben and Carl share \$240 in the ratio 2 : 3 : 5.
Calculate how much more Carl receives than Alex.

% ..... **[2]**

3 **(a)** Write the ratio 450 g to 1.2 kg in its simplest form.

..... **[2]**

(b) Anna, Ben and Chloe share \$420 in the ratio 3 : 4 : 5.
Calculate the amount Ben receives.

\$ ..... **[2]**

4 Alex, Ben and Charlie share \$840 in the ratio 2 : 3 : 5.

(a) Calculate the amount of money Ben receives.

\$ ..... **[2]**

(b) Express Charlie's share as a fraction of the total amount, giving your answer in its simplest form.

..... **[2]**

5 Alex, Ben and Chloe share \$120 in the ratio 3 : 4 : 5.

(a) Show that Chloe receives \$50.

[2]

(b) Work out the amount Alex receives.

\$ ..... **[2]**

(c) Write down the ratio of Ben's share to Chloe's share in its simplest form.

..... **[1]**

6 **(a)** Write the ratio 45 minutes to 2 hours in its simplest form.

..... **[2]**

(b) A piece of wood of length 3.6 m is cut into three pieces in the ratio 2 : 3 : 7.
Calculate the length, in cm, of the longest piece.

..... cm **[3]**

7 A sports club has m male members and f female members.
When 30 new male members join the club, the ratio of male members to female members becomes 3 : 1.
When, instead, 20 new female members join the original members, the ratio of male members to female members becomes 1 : 1.

(a) Write down two simultaneous equations in m and f.

.....

..... **[2]**

(b) Solve the equations to find the value of m and the value of f.

$m =$ ..... , $f =$ ..... **[3]**

(c) Find the original ratio of male members to female members in its simplest form.

..... **[1]**

8 A bag contains red balls, blue balls and green balls.
The ratio of red balls to blue balls is 3 : 4.
The ratio of blue balls to green balls is 5 : 2.

(a) Find the ratio of red balls to blue balls to green balls in its simplest form.

..... **[2]**

(b) There is a total of 129 balls in the bag.
Calculate the number of green balls.

..... **[2]**

(c) y blue balls and y green balls are removed from the bag.
The new ratio of blue balls to green balls is 3 : 1.
Work out the value of y.

$y =$ ..... **[3]**

9 A map is drawn to a scale of 1 : 50000.

(a) The length of a road on the map is 4.5 cm.
Calculate the actual length of the road in kilometres.

..... km **[2]**

(b) A lake has an actual area of 14.2 km^2.
Calculate the area of the lake on the map in cm^2.

..... cm^2 **[3]**

10 A map is drawn to a scale of 1 : 250 000.

(a) The distance between two towns on the map is 8.4 cm. Calculate the actual distance between the two towns in kilometres.

..... km **[3]**

(b) A forest has an actual area of 35 square kilometres.
Calculate the area of the forest on the map in square centimetres.

..... cm^2 **[3]**

11 A map is drawn to a scale of 1 : 250 000.

(a) The distance between two cities on the map is 16.4 cm.
Calculate the actual distance between the two cities in kilometres.

..... km **[2]**

(b) The actual area of a lake is 35 km^2.
Calculate the area of the lake on the map in square centimetres.

..... cm^2 **[3]**

(c) Another map is drawn such that a length of 5 cm represents an actual distance of 20 km.
Write the scale of this map in the form 1 : n.

1 : ..... **[1]**

12 **(a)** A map has a scale of 1 : 50000.
The actual distance between two towns is 14.5 km.
Calculate the distance, in cm, between the two towns on the map.

..... cm **[2]**

(b) An area of a forest on the map is 12.4 cm^2.
Calculate the actual area of the forest in km^2.

..... km^2 **[2]**

(c) A recipe for 8 people requires 240 g of flour and 150 g of butter.
Marco wants to make the recipe for 12 people.
He has 400 g of flour and 200 g of butter.
Determine if Marco has enough ingredients.
Show your working.

.....

..... **[3]**

13 A map is drawn to a scale of 1 : 250 000.

(a) The distance between two cities on the map is 16.8 cm.
Calculate the actual distance between the cities, in km.

..... km **[2]**

(b) A lake has an actual area of 56 km^2.
Calculate the area of the lake on the map, in cm^2.

..... cm^2 **[2]**

(c) On a different map of the same region, the area of the same lake is 14 cm^2.
Find the scale of this new map in the form 1 : n.

1 : ..... **[3]**

14 **(a)** The ratio of adults to children in a theater is 5 : 3.
There are 84 more adults than children.
Calculate the total number of people in the theater.

..... **[3]**

(b) A map is drawn to a scale of 1 : 25000.

(i) Two towns are 8.5 cm apart on the map.
Calculate the actual distance between the towns in kilometres.

..... km **[2]**

(ii) A forest has an actual area of 3.5 km^2.
Calculate the area of the forest on the map in square centimetres.

..... cm^2 **[3]**

15 In a factory, the number of male workers to female workers is in the ratio 5 : 3.
The ratio of male workers who work full-time to male workers who work part-time is 3 : 2.
The ratio of female workers who work full-time to female workers who work part-time is 4 : 1.

(a) Find the ratio of the total number of full-time workers to the total number of part-time workers in the factory.
Give your answer in its simplest form.

..... **[4]**

(b) There are 42 more full-time male workers than part-time female workers.
Calculate the total number of workers in the factory.

..... **[3]**

16 In a factory, the number of workers on the day shift and the number of workers on the night shift are in the ratio 5 : 3.

(a) There are 42 more workers on the day shift than on the night shift.
Calculate the total number of workers in the factory.

..... **[3]**

(b) The ratio of male workers to female workers on the day shift is 3 : 4.
The ratio of male workers to female workers on the night shift is 2 : 7.
Calculate the overall ratio of male workers to female workers in the factory.
Give your answer in its simplest form.

..... **[4]**

17 **(a)** In a club, the ratio of the number of boys to the number of girls is 3 : 5.
When 4 boys leave and 8 girls join the club, the new ratio of boys to girls becomes 1 : 3.
Calculate the original number of boys in the club.

..... **[4]**

(b) The ratio of the mass of object A to object B is 3 : 4.
The ratio of the mass of object B to object C is 5 : 6.
The mass of object C is 45 kg more than the mass of object A.
Calculate the total mass of the three objects.

..... kg **[3]**

18 Three partners, Alex, Ben and Chris, share a total profit of $\$P$. The ratio of Alex's share to Ben's share is 3 : 4. The ratio of Ben's share to Chris's share is 5 : 6.

(a) Find the ratio of Alex's share to Ben's share to Chris's share in its simplest form.

..... **[2]**

(b) Chris receives \$16 200 more than Alex. Calculate the value of P.

\$ ..... **[3]**

(c) Alex invests his share of the profit in a bank account that pays compound interest at a rate of 4.2% per year.
Calculate the value of Alex's investment at the end of 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**